Research highlights

LASER QUEST: HOW TO FIRE BURSTS OF NEUTRINOS

Physicists have proposed a type of laser that could emit powerful bursts of neutrinos instead of light. This could enable 'tabletop' studies of neutrinos — elementary particles that have no electrical charge — without expensive equipment such as accelerators and nuclear reactors.

In ordinary lasers, atoms are excited to a higher-energy state and stimulated to emit photons, or particles of light, in sync. In 'superradiant' lasers, atoms also emit photons in sync. But, unlike ordinary lasers, superradiant lasers do not rely on photons' 'bosonic' nature (a property that enables multiple photons to share identical quantum states). As a result, it should be possible to design a superradiant laser that emits neutrinos, which are not bosonic.

Benjamin Jones at the University of Texas at Arlington and Joseph Formaggio at the Massachusetts Institute of Technology in Cambridge calculated what would happen to a collection of around one million radioactive rubidium atoms that was suspended in a vacuum and cooled to extremely low temperatures. The authors found that the radioactive decay of the rubidium nucleus — which emits a neutrino — would occur 100,000 times faster than usual, producing laser-like bursts of neutrinos.

Phys. Rev. Lett. **135**, 111801 (2025)

HAVING A LOCAL HOST FAMILY HELPS REFUGEES THRIVE

Refugees hosted by local families integrate better into their new country than do those living in other types of accommodation.

Mathis Herpell, at the German Institute for Economic Research in Berlin, and his colleagues investigated the lives of Ukrainian refugees in Germany after the Russian invasion of Ukraine in 2022.

At the time, the large influx of displaced Ukrainians into Western Europe led to programmes in which residents opened their homes to them. The researchers surveyed refugees who were matched with local families through one of these programmes, as well as refugees who applied but were not matched.

Those who were hosted by local families reported greater social and psychological integration. As a result, they were able to form more ties with residents of their host country, and feel more connected to it, than were those living in public asylum centres or privately rented homes without a host. They also showed an improved capacity to handle essential tasks, such as seeing a doctor.

Refugees who reported the closest contact with their hosts were the most integrated, pointing to the value of these daily interactions.

Nature Hum. Behav. https://doi.org/p5p8 (2025)

A SUPERCOOL OUTFIT: GEL-FILLED GARMENT HELPS BEAT THE HEAT

A battery-powered garment embedded with squishy, water-laden gel could help people to work in extreme heat and humidity.

Yu Pei at the University of California, San Diego, and her colleagues created a garment from fabric covered with tile-shaped thermoelectric devices (TEDs) that use a voltage to create a temperature difference. When the authors ran an electrical current through the TEDs, the side facing the wearer's skin became cold, and the side facing away from the body became hot.

On the hot side of each TED was a layer of hydrogel, a material that contains large amounts of water. The heat from the TEDs raises the hydrogels' surface temperature to speed up evaporation while keeping the skin comfortably cool. Tested at temperatures of up to 55 °C and at humidity levels as high as 88%, the TEDs stayed below 35.8 °C on their cold side for more than six hours.

The garment is powered by a lightweight battery pack. Its cost is projected to be under US$20 when mass-produced, which the authors say could make the device widely accessible.

Cell Rep. Phys. Sci. **6**, 102816 (2025)

SONGS OF STRIPED MICE DISTINGUISH FRIEND FROM FOE

In the African desert, wild mice 'sing' tunes that don't travel far but help the animals to tell neighbours from strangers, researchers have found.

Indigenous to southern Africa, striped mice (*Rhabdomys pumilio*, pictured) live in groups of up to 30, gathering in nests at night but foraging alone by day. Léo Perrier at the University of Saint-Etienne in France and his colleagues tracked four groups of striped mice and set up 23 sound recorders: some at nests, others in the middle of the animals' territories or in border areas where different groups meet.

Over 12 days, the researchers recorded thousands of ultrasonic calls, which travel less than 2 metres and are too high-pitched for humans to hear. The mice 'sang' throughout their territory, changing their calls at the borders. Each group of mice had its own vocal signature.

Playback experiments confirmed that the mice can distinguish friends from strangers: hearing an outsider triggered retreat, whereas hearing familiar calls didn't. This ultrasonic chatter helps the mice to communicate across their territory and interact with neighbours and unfamiliar mice, the authors say.

Curr. Biol. https://doi.org/p5p9 (2025)

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